

Geospatial Analysis Portfolio

Katherine M. Yut

Wednesday, April 15, 2026

Contents

Cover Letter	2
References	5
Sample Work	6
Grassroots Carbon	6
Bureau of Economic Geology	7
Internships	10
E&J Gallo Winery (2020)	10
NASA DEVELOP (2019)	12

Cover Letter

Dear NASA,

I am writing to express my interest in the Program Scientist position at NASA and outline my qualifications, which firmly position me as an outstanding potential team member. With a background in data wrangling, geospatial analysis, and the art of data visualization, I possess strong problem-solving abilities coupled with a knack for coding, making me an ideal candidate for data-driven endeavors.

During my tenure as a GIS & Energy Data Analyst at the University of Texas, Bureau of Economic Geology, I have honed comprehensive skills in GIS, coding, and effective communication. My track record includes successfully utilizing tools such as ArcMap and R to perform intricate subsurface research, develop peer-reviewed literature, and present findings.

My experience as a Viticulture Research GIS Intern at E. & J. Gallo Winery has equipped me with the proficiency to manage large datasets, automate satellite data acquisition, and employ statistical techniques to drive decision-making. Collaborating with cross-functional teams, I contributed to optimizing wine grape yield and quality through informed irrigation strategies, solidifying my dedication to translating data into impactful solutions.

While a remote sensing intern with NASA DEVELOP, I conducted thorough literature reviews, collected training data, deployed classification models, and performed site suitability analyses. My role demanded a keen eye for design as I created compelling visuals while ensuring adherence to stringent style guidelines. I excelled at developing workflows using ArcGIS, ENVI, Idrisi Terrset, and Google Earth Engine despite unforeseen challenges. Thriving in a collaborative environment, my team and I effectively communicated complex scientific concepts to both technical and non-technical audiences.

My educational background, consisting of a Master of Arts in Economics and Big Data, a Bachelor of Arts in Economics and Big Data, and a Bachelor of Science in Geographic Information Science, demonstrates my curiosity, rapid learning capability, and unwavering commitment to acquiring expertise across domains. My skill set includes RStudio, R, ESRI ArcMap, QGIS, Google Earth Engine, ENVI, GitHub, Java, JavaScript, SQL, and Python.

Driven by a genuine passion for data analysis, I am eager to contribute my insights, skills, and efficiency to the team. Thank you for considering my application. I look forward to the opportunity to discuss how my experiences align with your vision during an interview.

Best,



Katherine (Katy) Yut

Katherine M. Yut

Geospatial Data Engineer

Expert in geospatial analysis, statistics, and programming using R and Python. Passionate about data visualization. Seeking to solve interesting environmental problems. Meticulous, intrinsically motivated, and a quick learner.

Work History

Grassroots Carbon, San Antonio, TX (hybrid)

Data Engineer 2025-08 – current

Building ETL/ELT data pipelines using Python, GitHub, and Docker containerization.

Skills: pydantic, Streamlit, cron, GitHub Actions, CI/CD.

Grassroots Carbon, San Antonio, TX (hybrid)

Carbon Technical Manager 2024-08 – 2025-08

Automated rancher eligibility screening processes using R shiny and Python, provided geospatial data analysis using QGIS, and supported the navigation of carbon standards.

University of Texas, Bureau of Economic Geology, Austin, TX

GIS & Energy Data Analyst 2022-05 – 2024-07

Leveraged R to clean and manipulate data tables, solve problems, and automate processes; utilized ArcMap to aid in oil & gas research publications; and published research findings as papers, posters, and presentations.

Education

Master of Arts: Economics and Big Data

Bachelor of Arts: Economics and Big Data

Bachelor of Science: Geographic Information Science

University of Oklahoma (Norman, OK)

2016-08 – 2021-05, *Summa cum laude*

Research

KM Smye, **K Yut**, RC Reedy, BR Scanlon, JP Nicot, P Hennings (2024). Challenges with managing unconventional water production and disposal in the Permian Basin. AAPG Bulletin 108 (12), 2215-2240.

Peng, S., Maraggi, LMR., Bhattacharya, S., **Yut, K.**, McMahon, T., Haddad, M. (2023). Feasibility of CO2 storage in depleted unconventional oil and gas reservoirs: Capacity, microscale mechanism, injectivity, fault stability, and monitoring. URTEC 13–15 June 2023.

M Bull, T Francis, **K Yut**, K Weber, J Spruce, C Jarnevich. West Virginia Ecological Forecasting. NASA DEVELOP National Program, 2019.

Kelley, S., **Yut, K.**, Kulkarni, R., & Gaffin, D. (2019). Avoidance of rosemary oil by scorpions. Journal of Arachnology.

Contact

katy.yut@gmail.com

(972) 757-4564

linkedin.com/in/katherine-yut/

Skills

Expert

RStudio, R, R shiny:

Writing functions, making packages, documenting code, creating interactive html servers.

ESRI ArcGIS Pro, QGIS:

Geospatial analyses, coordinate system management, map-making.

Creating timely deliverables and communicating effectively

Curiosity, organization, multitasking, and focus

Advanced

Python

GitHub

Big Data statistical processing, analysis, and modeling

Intermediate

SQL

Docker containerization

Google Earth Engine

ENVI

References

The following individuals are happy to vouch for my character and data analysis capabilities:

1. Chip Thompson Grassroots Carbon | Chief Financial Officer
chip@buildgrassroots.com | (210) 687-3594
2. Katie Smye
Bureau of Economic Geology | Research Associate
katie.smye@beg.utexas.edu | (512)-471-6775
3. Brent Sams
E. & J. Gallo Winery | Research Scientist
sams.brents@gmail.com | (209) 568-9317

Sample Work

Grassroots Carbon

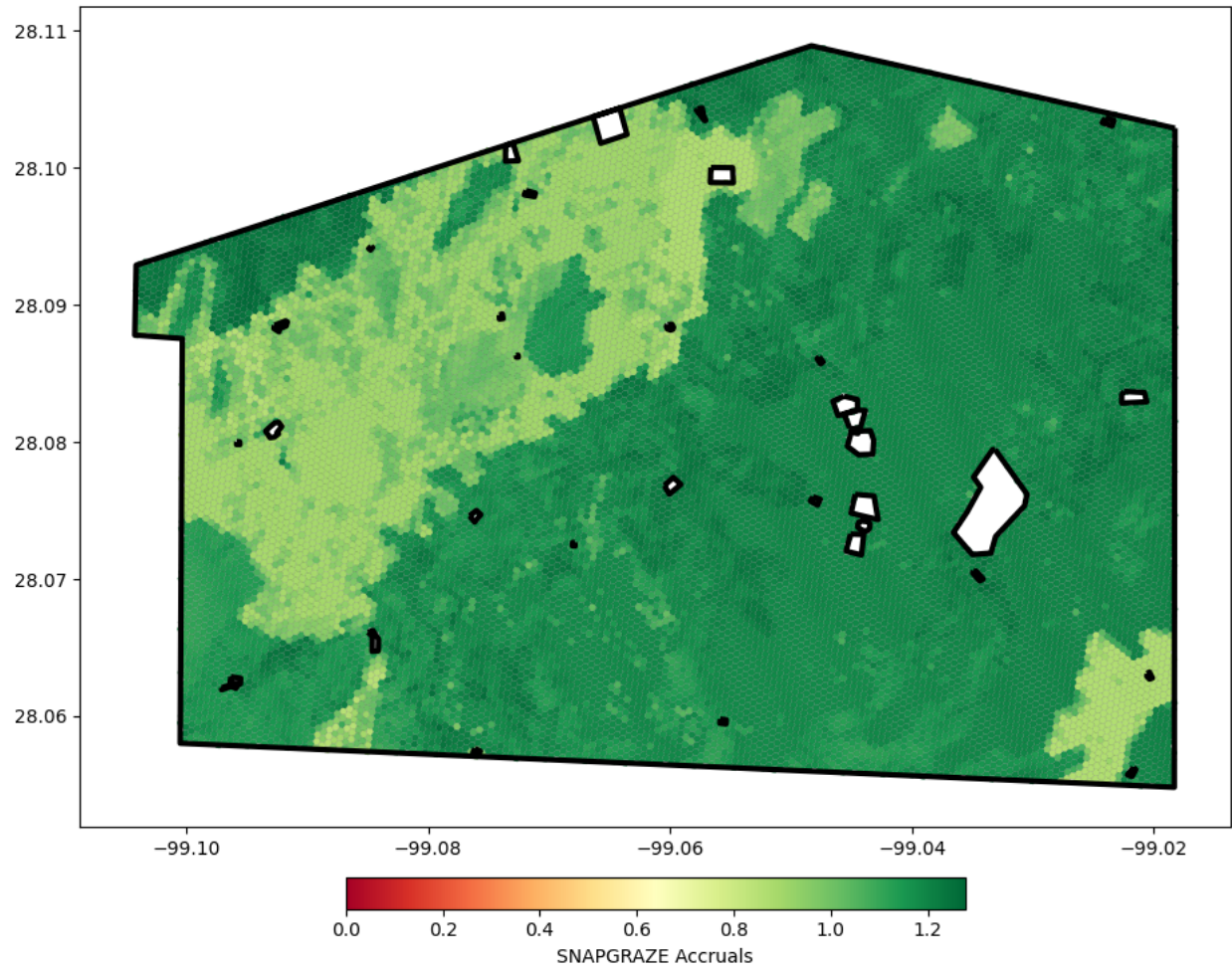


Figure 1: Heatmap of SNAPGRAZE model outputs, showing SOC accrual potential on a ranch in Texas. Created in Python using matplotlib and uber H3 hexagons.

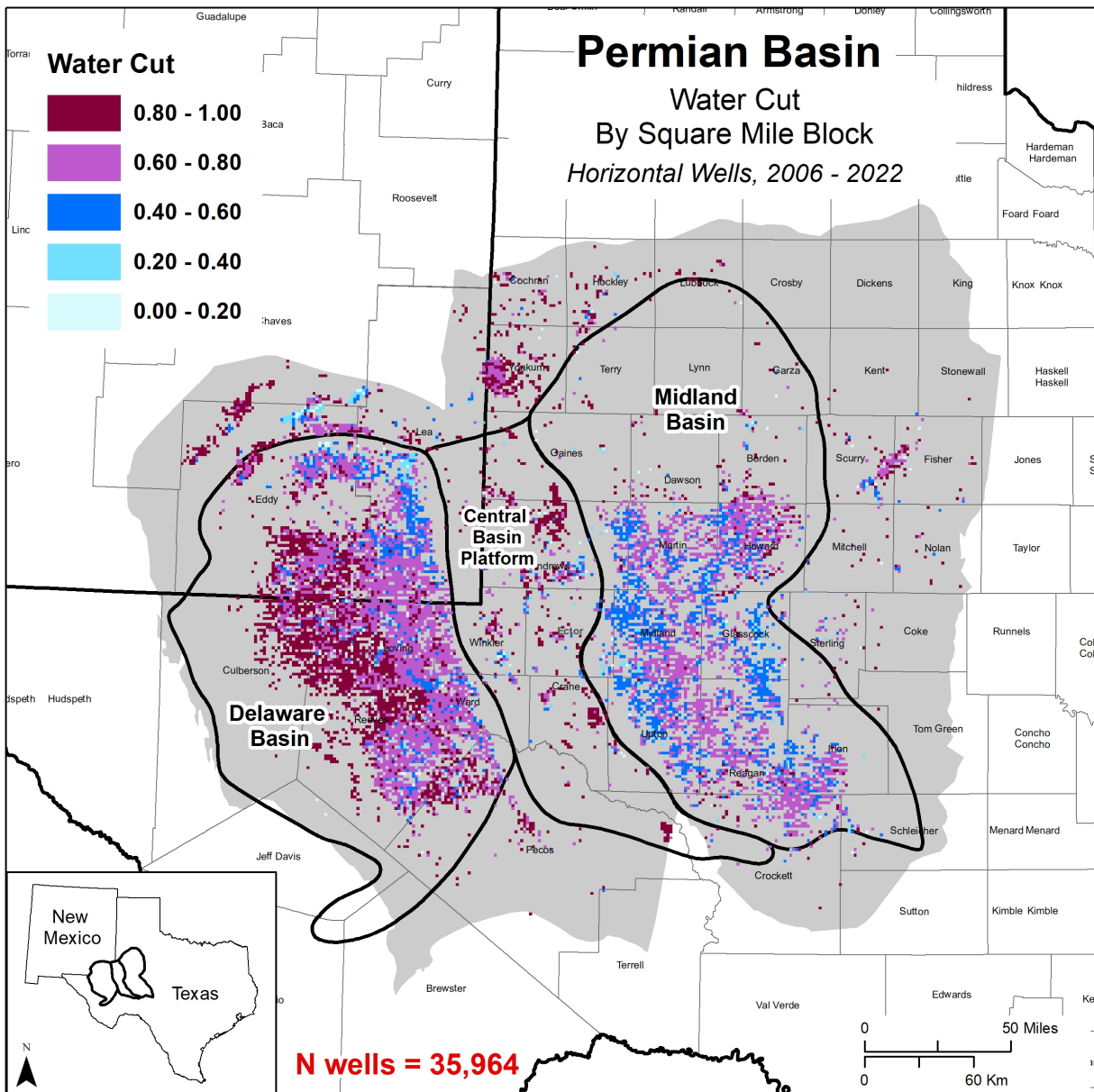


Figure 2: Map of Water Cut ($\text{Produced Water} / \text{Produced Oil} + \text{Produced Water}$) for all horizontal wells in the Permian Basin, aggregated on square mile blocks. Created in ArcMap using data from IHS Enerdeq.

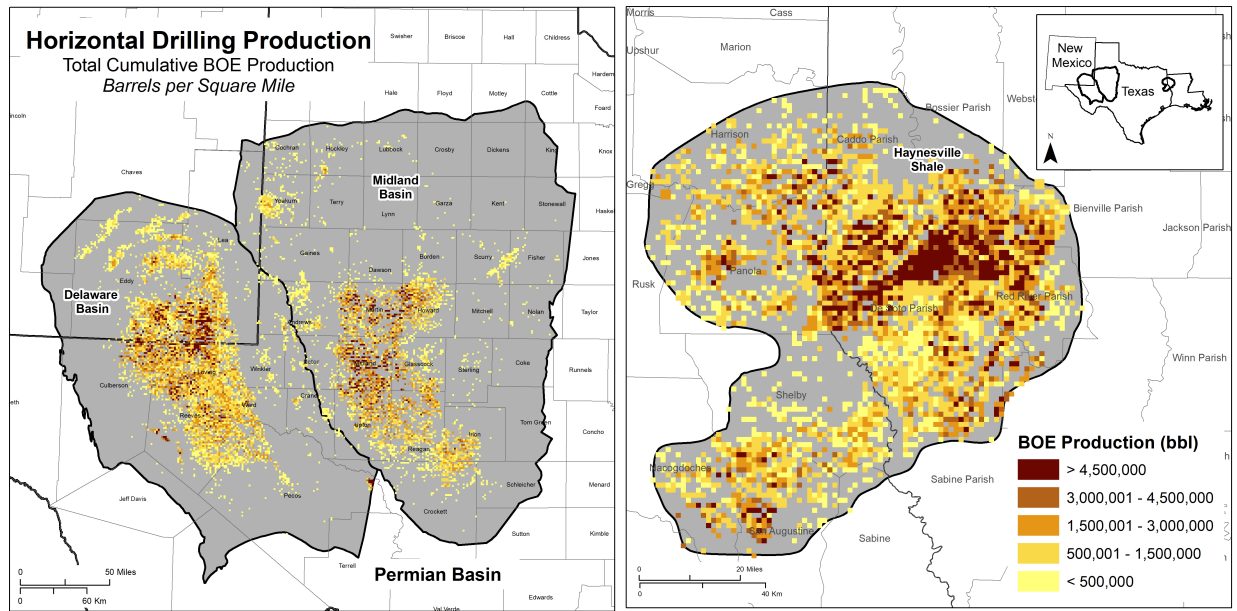


Figure 3: Map of total cumulative production of oil and gas production, in barrels of oil equivalent, for all horizontal wells in Delaware, Midland, and Haynesville, aggregated on square mile blocks. Created in ArcMap using data from IHS Enerdeq.

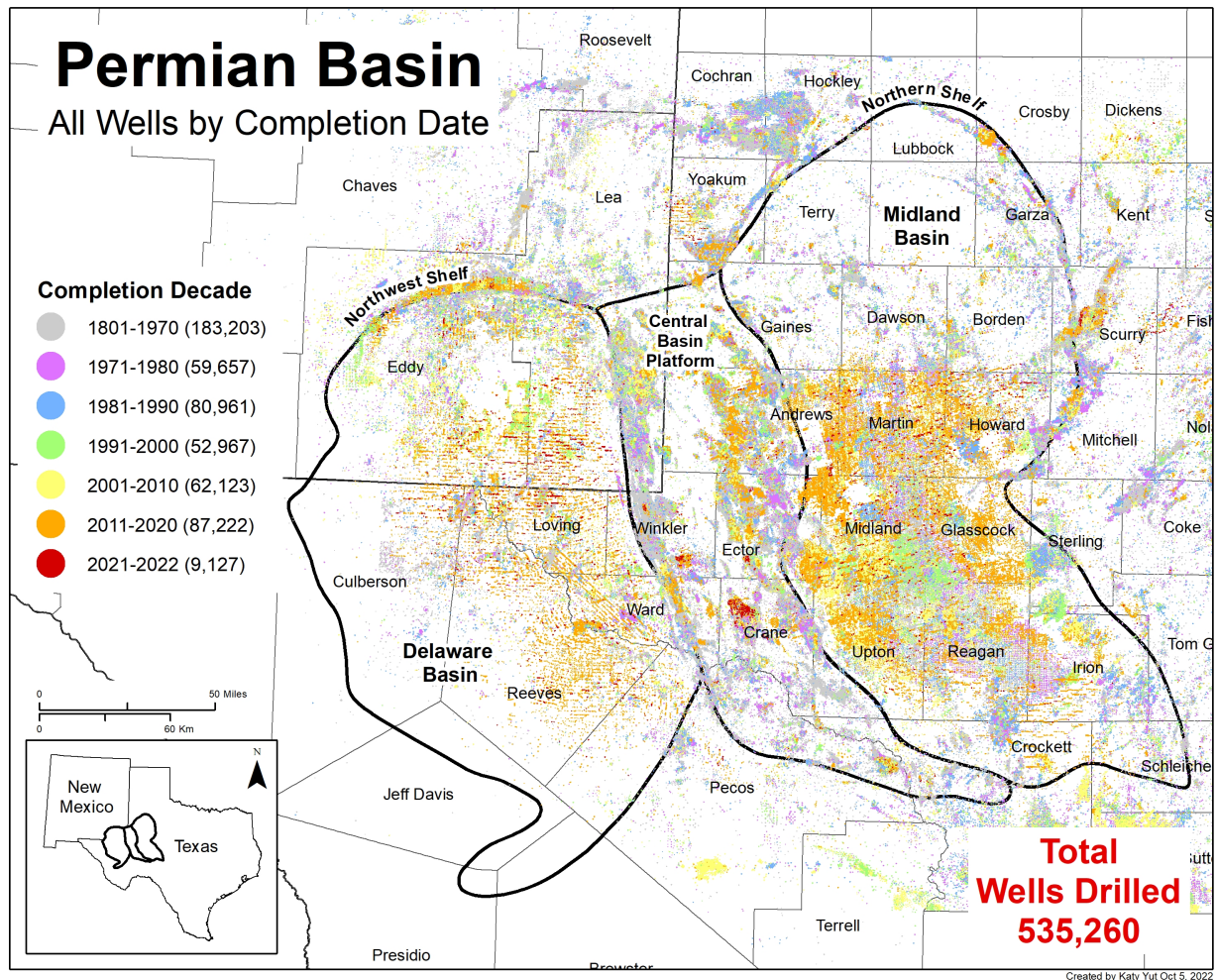


Figure 4: Map of all wells drilled in the Permian Basin region, colored by decade completed. Created in ArcMap using data from IHS Enerdeq.

Internships

E&J Gallo Winery (2020)

World Grape Production

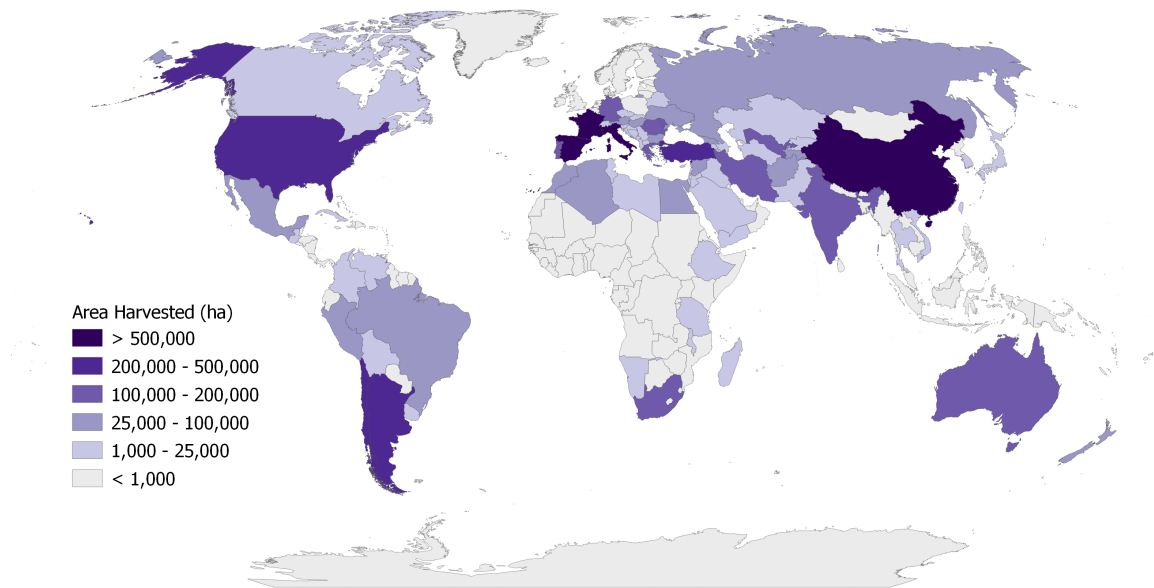


Figure 5: World map of grape production by country. Map requested and produced within one business day. Created in QGIS.

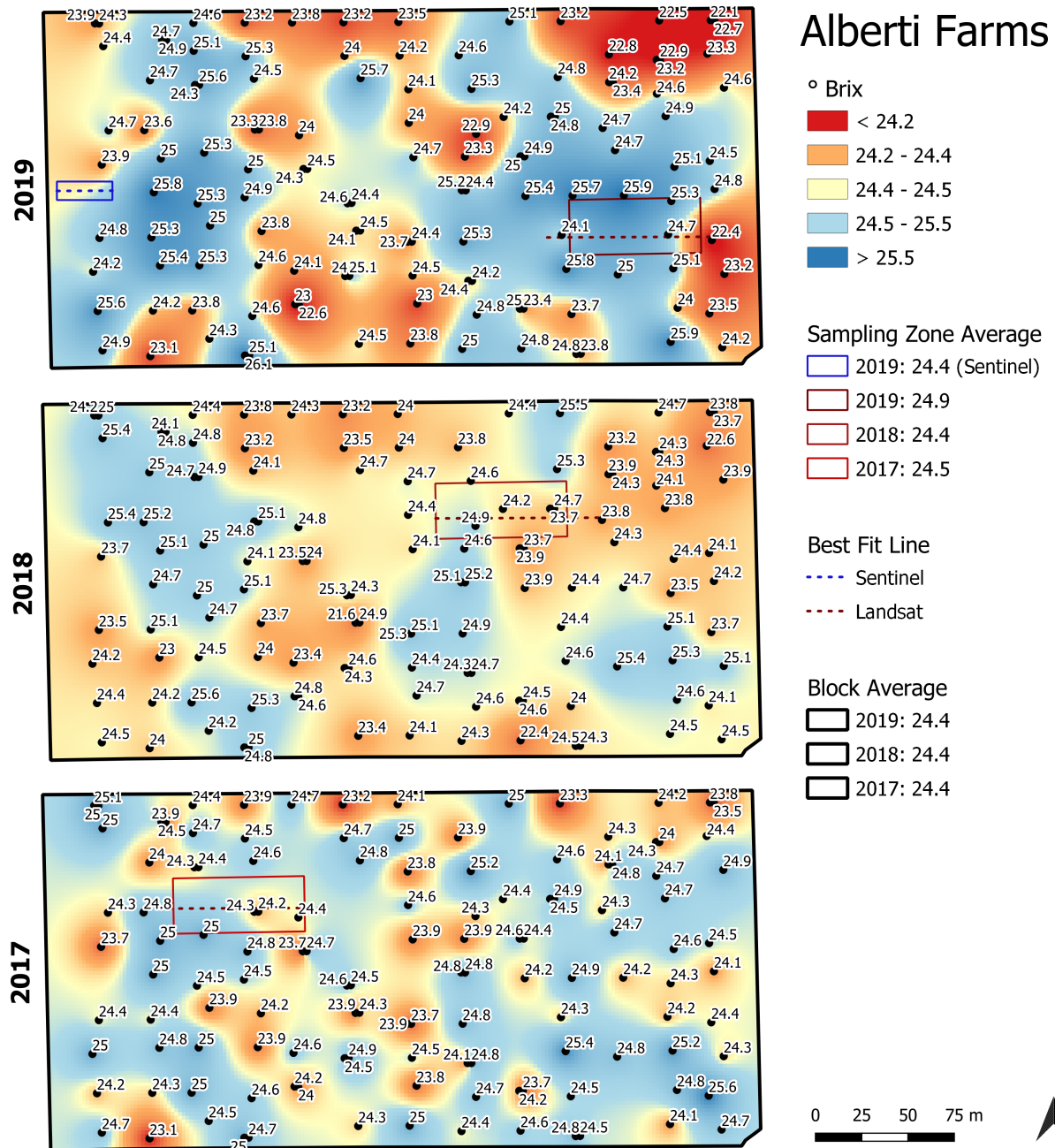


Figure 6: Kriging interpolated Brix (i.e., grape sugar content) values for 2017, 2018, and 2019 across a single block with sampling algorithm solutions for both Landsat and Sentinel imagery. Created in QGIS.

NASA DEVELOP (2019)

Forecasting Red Spruce Restoration Using NASA Earth Observations to Support Decision Making in the USFS Monongahela National Forest

ABSTRACT: In the Monongahela National Forest (MNF), situated in the Allegheny Highlands of West Virginia, extensive logging and mining practices have significantly altered the structure and composition of flora and fauna over the past two centuries. To aid red spruce restoration, this study mapped current and historical stands and identified non-native stands. . . [Read full abstract here.](#)

Monongahela National Forest Red Spruce LULC 2018 to 2040

National Aeronautics and Space Administration

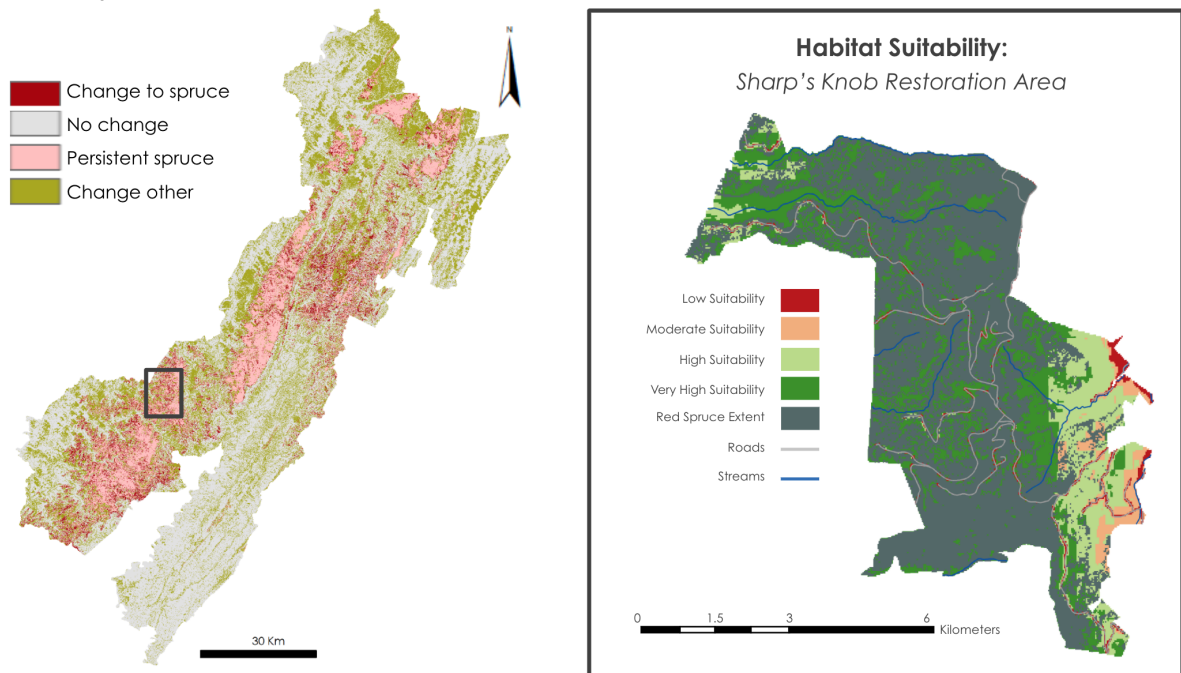


Figure 7: Land classification change using Classification Tree Algorithm in Idrisi Terrset (left). Site suitability analysis created in ArcMap (right).